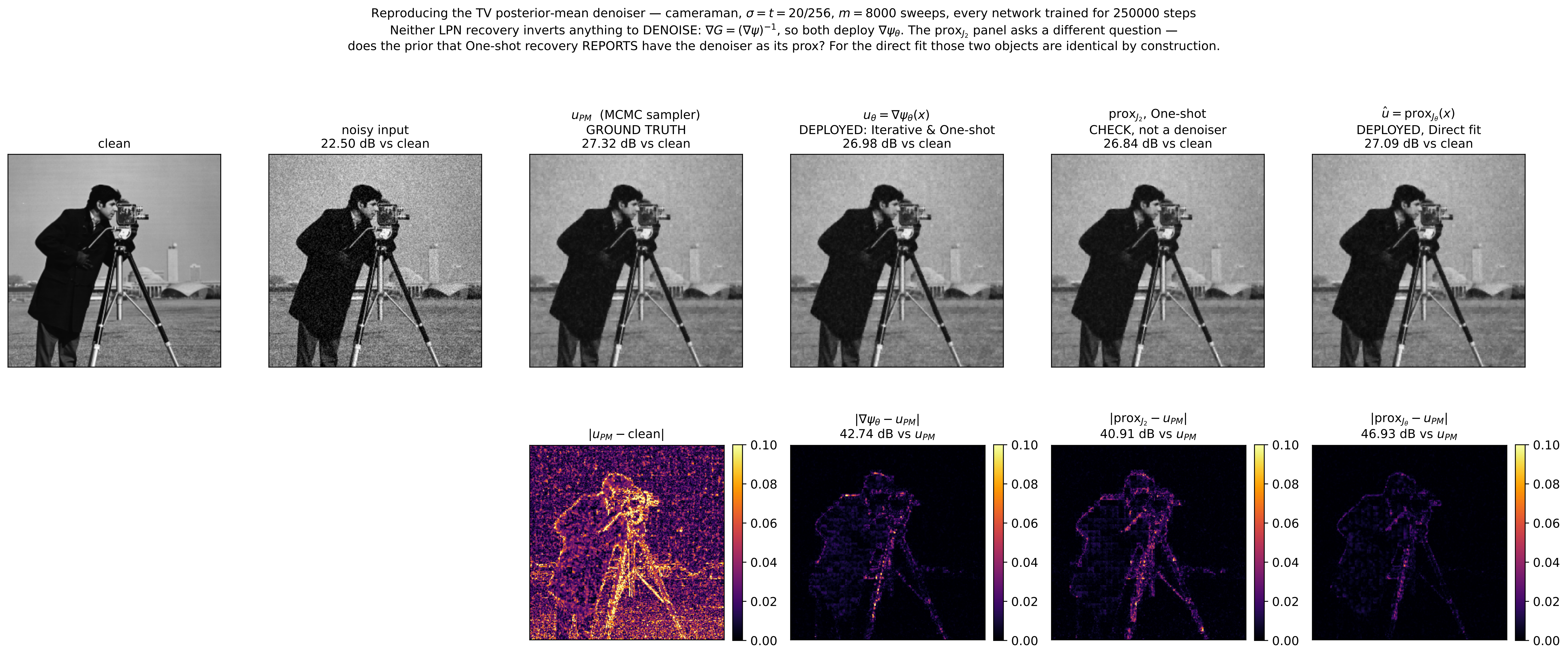
Reproducing the TV posterior-mean denoiser — cameraman, $\sigma = t = 20/256$, $m = 8000$ sweeps, every network trained for 250000 steps
Neither LPN recovery inverts anything to DENOISE: $\nabla G = (\nabla\psi)^{-1}$, so both deploy $\nabla\psi_\theta$. The prox_{J_2} panel asks a different question —
does the prior that One-shot recovery REPORTS have the denoiser as its prox? For the direct fit those two objects are identical by construction.
clean
noisy input
22.50 dB vs clean
u_{PM} (MCMC sampler)
GROUND TRUTH
27.32 dB vs clean
$u_\theta = \nabla\psi_\theta(x)$
DEPLOYED: Iterative & One-shot
26.98 dB vs clean
prox_{J_2}, One-shot
CHECK, not a denoiser
26.84 dB vs clean
$\hat{u} = \mathrm{prox}_{J_\theta}(x)$
DEPLOYED, Direct fit
27.09 dB vs clean
$|u_{PM} - \mathrm{clean}|$
$|\nabla\psi_\theta - u_{PM}|$
42.74 dB vs u_{PM}
$|\mathrm{prox}_{J_2} - u_{PM}|$
40.91 dB vs u_{PM}
$|\mathrm{prox}_{J_\theta} - u_{PM}|$
46.93 dB vs u_{PM}
0.10
0.08
0.06
0.04
0.02
0.00